

Rugveda Desai

LinkedIn: rugvedadesai

GitHub: <https://github.com/rugdesai>

Email : desairugveda@gmail.com

Mobile : 8792847106

EDUCATION

Manipal Institute of Technology, Bangalore

B.Tech in Computer Science and Data Science; CGPA: 8.23

Bangalore, Karnataka

July 2024 – May 2028

EXPERIENCE

Cognifyz Technologies

Data Science Intern

Maharashtra

Summer 2025

- Conducted data preprocessing, feature engineering, EDA, and visualization on structured datasets using Python data science libraries.

Manipal Institute of Technology

Research Intern

Bangalore, Karnataka

March 2025 – Present

- Researched LLM adversarial attacks, prompt injection, and model robustness while contributing to survey publications on foundation model security and AI-driven forecasting methods.

PUBLICATIONS & RESEARCH HIGHLIGHTS

Adversarial Attacks on Foundation Models: Accepted for peer-reviewed oral presentation at the Second International Conference on Artificial Intelligence, Computation, Communication and Network Security, University of Wollongong Dubai, UAE.

Wellora: Selected under **Yukti Innovation (Government of India)** for presentation at the **IIC Regional Meet**, Bengaluru.

PROJECTS

CollabNote: Real-Time Collaborative Note Editor

June 2025

React, TypeScript, Node.js, Socket.io, PostgreSQL, Redis, Docker

- Built a real-time collaborative editor with Socket.io supporting multi-user editing, live cursors, and user presence tracking.
- Optimized performance with debounced auto-save and Redis pub/sub architecture, reducing database writes and enabling horizontal scaling
- Implemented JWT authentication, PostgreSQL data models, and Dockerized full-stack deployment on Railway.

Wellora: Biomedical Mental Health Platform

April 2025 – July 2025

Research Integration, Python, Dart, ML, System Design

- Developed a biomedical mental health platform integrating ML-based features with a modular full-stack application architecture.
- Translated clinical research requirements into scalable technical specifications, workflows, and system components while collaborating with stakeholders.

PrepPal — AI Mock Interview Platform

In Development

Next.js, FastAPI, OpenAI API, PostgreSQL

- Developing an AI-powered mock interview tool that generates role-specific questions and streams structured feedback using the OpenAI API with FastAPI's async streaming support
- Designing role-aware evaluation rubrics so feedback criteria adapt based on the selected job role (SWE, PM, Data Analyst)

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, SQL, HTML/CSS

Frontend: React, Next.js, Tailwind CSS, Context API, React Hooks

Backend: Node.js, Express.js, FastAPI, RESTful APIs, JWT Authentication, Middleware Design

Databases: PostgreSQL, MongoDB, Redis (Pub/Sub)

Real-Time Systems: Socket.io (WebSockets), Repository Pattern

DevOps Tools: Docker, Docker Compose, Git, GitHub, Postman, Railway/Render Deployment

AI/LLM: OpenAI API Integration, Prompt Engineering, LangChain Basics, RAG Fundamentals

Data Science & Visualization: Pandas, NumPy, Matplotlib, Seaborn, OpenCV, Streamlit

LEADERSHIP & EXTRACURRICULARS

President, CodeX Tech Club: Led research-oriented technical initiatives and mentored peers on SWE project development

Organizing Committee Tech Solstice : a campus-wide technical event with 2000+ participants